

Fourier-Frame Self-Affine Measures Are Self-Affine Tiles

fa_banach_001 / agent_lane_00

17 August 2026

Status: candidate full affirmative resolution of Q2 under the paper's standard equal-weight, common-matrix convention.

1 The question and the precise result

Lai proves that a compactly supported absolutely continuous probability measure $d\mu = \varphi dx$ admits a Fourier frame exactly when its density is bounded above and bounded away from zero almost everywhere on its support [1, Theorem 1.1]. After applying this criterion to equal-weight self-similar measures on the line, the paper asks whether the analogous conclusion holds for self-affine measures on $\mathbb{R}^d$ [1, p. 13, Q2].

self-affine) measure on $\mathbb{R}^n$ that is absolutely continuous with density bounded above and below almost everywhere on its support. It is conjectured that Theorem 4.2 will hold more generally:

Q2: Is it true that the only absolutely continuous self-affine measures on $\mathbb{R}^n$ admitting a Fourier frame is the characteristic function of self-affine tiles?

This paper focuses only on the absolutely continuous measures, the question be-

We use the natural higher-dimensional version of the class treated in the paper. Let R be an expansive invertible real $d \times d$ matrix, let $B \subset \mathbb{R}^d$ consist of N distinct digits, and set

$$\tau_b(x) = R^{-1}(x + b) \quad (b \in B).$$

Let K be the attractor and μ the equal-weight invariant probability:

$$K = \bigcup_{b \in B} \tau_b(K), \quad \mu(E) = \frac{1}{N} \sum_{b \in B} \mu(\tau_b^{-1}E). \quad (1)$$

Write $q = |\det R|$.

Theorem 1. *Suppose that μ is absolutely continuous and that $L^2(K, d\mu)$ admits a Fourier frame. Then*

$$q = N, \quad \mu = \frac{1}{\mathcal{L}^d(K)} \mathcal{L}^d \upharpoonright_K,$$

and the sets $\{\tau_b(K) : b \in B\}$ partition K up to null sets. Consequently $RK = K + B$ is a self-affine tile in the standard sense of Lagarias–Wang [2]; in particular, K tiles $\mathbb{R}^d$ by translations. Thus Q2 has an affirmative answer in this class, with the harmless probability-normalization correction that the density is $\mathcal{L}^d(K)^{-1} \chi_K$, not literally χ_K unless $\mathcal{L}^d(K) = 1$.

2 Proof intuition

Positive Lebesgue measure first gives the easy volume inequality $q \leq N$. For the reverse inequality, choose a generic linear functional. At every digit level it selects a unique minimizing digit, producing one address whose point x_* is exposed not merely in K , but in a unique cylinder at every depth. Near x_* , the iterated density equation therefore has only one surviving summand. Each refinement multiplies that summand by q/N . The Fourier-frame criterion bounds the density above and below, so this factor cannot decay through arbitrarily many refinements; hence $q \geq N$. Equality forces no overlap in measure, and uniqueness of the invariant probability then identifies μ with normalized Lebesgue measure on K .

3 A cylinder isolated at every depth

Every point of K has an address and

$$K = \left\{ \sum_{j=1}^{\infty} R^{-j} c_j : c_j \in B \right\}. \quad (2)$$

For a word $w = (c_1, \dots, c_n)$ write $\tau_w = \tau_{c_1} \circ \dots \circ \tau_{c_n}$ and $K_w = \tau_w(K)$.

Lemma 1. *There are digits $b_1, b_2, \dots$ and a point $x_* = \sum_{j \geq 1} R^{-j} b_j \in K$ such that, for every n ,*

$$x_* \in K_{(b_1, \dots, b_n)} \quad \text{and} \quad x_* \notin K_w \quad \text{for every other word } w \in B^n.$$

Consequently, for every n there is $r_n > 0$ such that

$$K \cap B(x_*, r_n) \subset K_{(b_1, \dots, b_n)} \setminus \bigcup_{w \neq (b_1, \dots, b_n)} K_w. \quad (3)$$

Proof. Choose $u \in \mathbb{R}^d$ outside the countable union of proper hyperplanes

$$\bigcup_{j \geq 1} \bigcup_{b \neq b'} \{u : \langle u, R^{-j}(b - b') \rangle = 0\}.$$

For each j , let b_j be the unique digit minimizing $\langle u, R^{-j}b \rangle$ over $b \in B$. The series defining x_* converges because R is expansive. If (c_j) is any other address, then term by term

$$\langle u, R^{-j}(c_j - b_j) \rangle \geq 0,$$

and the inequality is strict at the first index where $c_j \neq b_j$. The absolutely convergent sum is therefore strictly positive. Hence x_* is the unique minimizer of $x \mapsto \langle u, x \rangle$ on K .

More is true. If a depth- n word w differs from $(b_1, \dots, b_n)$, then every address beginning with w has a strict positive contribution at one of its first n levels and nonnegative contributions thereafter. Thus $x_* \notin K_w$. The union of the finitely many other compact cylinders is compact and omits x_* , so its distance from x_* is positive. A sufficiently small ball gives (3). $\square$

4 Proof of the theorem

Proof of Theorem 1. By Lai's Fourier-frame criterion there are constants $0 < m \leq M < \infty$ such that, after extending the density by zero off K ,

$$m \leq \varphi(x) \leq M \quad \text{for almost every } x \in K. \quad (4)$$

In particular $\mathcal{L}^d(K) > 0$. Taking Lebesgue measure in $K = \bigcup_{b \in B} \tau_b(K)$ gives

$$\mathcal{L}^d(K) \leq \sum_{b \in B} \mathcal{L}^d(\tau_b(K)) = \frac{N}{q} \mathcal{L}^d(K),$$

and hence

$$q \leq N. \quad (5)$$

Changing variables in (1) gives the density refinement identity

$$\varphi(x) = \frac{q}{N} \sum_{b \in B} \varphi(Rx - b) \quad \text{for almost every } x \in \mathbb{R}^d. \quad (6)$$

Its n -fold iterate is

$$\varphi(x) = \left(\frac{q}{N}\right)^n \sum_{w \in B^n} \varphi(\tau_w^{-1}x) \quad \text{for almost every } x. \quad (7)$$

Fix n and use Lemma 1. Put $E_n = K \cap B(x_*, r_n)$. Since every positive-weight IFS measure has support equal to its attractor, $x_* \in \text{supp } \mu$ and $\mu(E_n) > 0$. Absolute continuity then implies $\mathcal{L}^d(E_n) > 0$. For $x \in E_n$, only the selected cylinder can contribute to (7); all other inverse images lie outside K . Combining (4) and (7), and discarding the union of the relevant null exceptional sets, yields on a positive-measure subset of E_n

$$m \leq \varphi(x) = \left(\frac{q}{N}\right)^n \varphi(\tau_{(b_1, \dots, b_n)}^{-1}x) \leq \left(\frac{q}{N}\right)^n M.$$

This holds for every n . If $q < N$, the right side is eventually smaller than m , a contradiction. Thus $q \geq N$, which together with (5) proves $q = N$.

Now the finite cover in (1) satisfies

$$\mathcal{L}^d(K) = \sum_{b \in B} \mathcal{L}^d(\tau_b(K)).$$

Therefore its pieces overlap only on a null set. Define normalized Lebesgue measure on K by $\nu(E) = \mathcal{L}^d(E \cap K) / \mathcal{L}^d(K)$. For every Borel set E ,

$$\begin{aligned} \frac{1}{N} \sum_{b \in B} \nu(\tau_b^{-1}E) &= \frac{q}{N \mathcal{L}^d(K)} \sum_{b \in B} \mathcal{L}^d(E \cap \tau_b(K)) \\ &= \frac{\mathcal{L}^d(E \cap K)}{\mathcal{L}^d(K)} = \nu(E), \end{aligned}$$

where the second equality uses $q = N$ and the almost-everywhere partition. Thus ν satisfies the same invariant equation as μ . Uniqueness of the invariant probability for a contractive IFS gives $\mu = \nu$.

Finally $RK = K + B$, K has positive measure, and B has $N = q = |\det R|$ digits. This is precisely the self-affine-tile condition of Lagarias–Wang; their tiling theorem supplies a translation tiling of $\mathbb{R}^d$. This completes the proof. $\square$

Checks, novelty, and scope

The exposed-address argument does not assume an open set condition, a lattice digit set, diagonalizability, or any ordering of $\mathbb{R}^d$. Its only geometric input is that an expansive matrix makes (2) converge. The measure-theoretic step is also insensitive to boundary points: support gives $\mu(E_n) > 0$, and absolute continuity gives $\mathcal{L}^d(E_n) > 0$.

The word “self-affine measure” is not separately defined around Q2. This packet therefore retains the equal weights and common linear part of the self-similar class explicitly treated immediately before the

question, while allowing a general expansive matrix and arbitrary dimension. Unequal weights or an IFS with varying linear parts are not claimed here; in those broader conventions, the factor along the isolated address is no longer $(q/N)^n$, so a genuinely new argument would be needed.

A bounded literature search used the run indexes and arXiv/web queries for the source identifier, exact question, self-affine Fourier frames, bounded self-affine densities, and the exposed-address/determinant mechanism. It found later work on uniformity and frame-spectral self-affine measures, but no paper stating this resolution of Q2. The tile terminology and final tiling implication are classical [2]. Novelty is therefore provisional pending expert review.

References

- [1] C.-K. Lai, *On Fourier frame of absolutely continuous measures*, Journal of Functional Analysis 261 (2011), 2877–2889; arXiv:1103.4691v2.
- [2] J. C. Lagarias and Y. Wang, *Self-Affine Tiles in $\mathbb{R}^n$* , Advances in Mathematics 121 (1996), 21–49.